

1 1. Time for CPU Conversion

```
height = img.shape[0]
width = img.shape[1]

cpu_gray_image = [
    [0 for _ in range(width)]
    for _ in range(height)
]

start = time.time()

for y in range(height):
    for x in range(width):
        r = int(img[y, x, 0])
        g = int(img[y, x, 1])
        b = int(img[y, x, 2])
        gray = (r + g + b) // 3
        cpu_gray_image[y][x] = gray

end = time.time()

print(f"CPU time: {end - start:.6f} seconds")

-> Results: CPU time: 0.266899 seconds
```

2 2. Time for GPU Conversion

```
@cuda.jit
def grayscale_kernel(input_img, output_img):
    tid = cuda.threadIdx.x + cuda.blockIdx.x * cuda.blockDim.x
    g = np.uint8((input_img[tid, 0] + input_img[tid, 1] + input_img[tid, 2]) / 3)
    output_img[tid, 0] = output_img[tid, 1] = output_img[tid, 2] = g

original_shape = img.shape
img_flat = img.reshape(-1, 3).astype(np.uint8)

d_img = cuda.to_device(img_flat)
d_grayscale_img = cuda.device_array(
    img_flat.shape,
    dtype=np.uint8
)
pixelCount = img_flat.shape[0]
blockSize = 256
```

```

gridSize = (pixelCount + blockSize - 1) // blockSize
start = time.time()
grayscale_kernel[gridSize, blockSize](d_img, d_grayscale_img)
cuda.synchronize()
end = time.time()
grayscale_img = d_grayscale_img.copy_to_host()
print(f"GPU time: {end - start:.6f} seconds")

```

Explanation:

1st: Flatten the image and send it to gpu
 2nd: Create a device array to store the result
 3rd: Calculate the grid size and block size
 4th: Call the kernel function and synchronize
 5th: Copy the result back to host and reshape it to original shape
 -> Results: GPU time: 0.056622 seconds

3 3. Block size vs Time

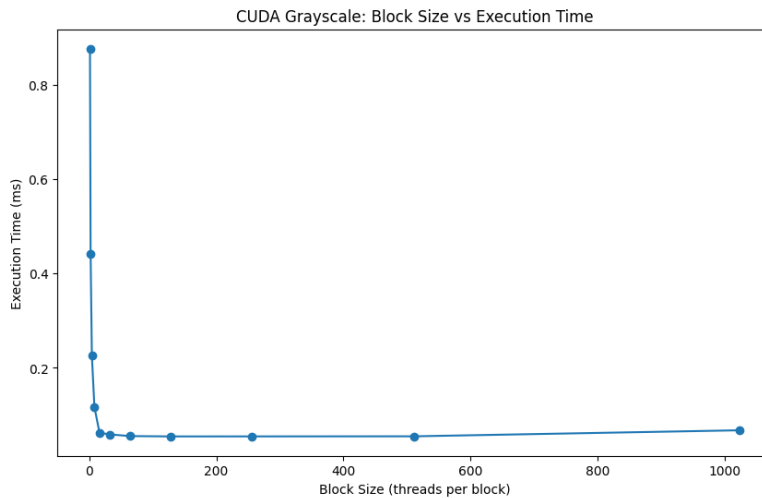


Figure 1: Block size vs Time

- Increase the block size, the time will decrease. The optimal block size is 256.